Artificial Intelligence in Microfinance and Financial Inclusion: Applications, Issues, and Future Directions

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Abstract

Artificial intelligence (AI) is emerging as a transformative force in microfinance and financial inclusion, addressing long-standing barriers such as credit invisibility, high operational costs, and limited access to formal financial services. This paper systematically examines AI applications across key financial domains (payments, savings, lending, insurance, investments) highlighting how machine learning, natural language processing, and generative AI are enabling innovative solutions tailored to the needs of marginalized populations. Drawing on contemporary research and case studies from the Global South, the analysis demonstrates AI's potential to democratize financial services through alternative credit scoring, automated underwriting, and adaptive tools.

However, the deployment of AI also presents significant challenges, including algorithmic bias, proxy discrimination, privacy violations, and the risk of exacerbating digital divides. The paper underscores the need for robust governance frameworks, ethical oversight, and inclusive policies to mitigate these risks and ensure that AI-driven financial inclusion serves the most vulnerable without creating new forms of exclusion. Future directions include advancing fairness-aware AI, improving transparency, and fostering cross-sector collaboration to align technological innovation with social justice and human dignity.

Keywords: Artificial Intelligence, Microfinance, Financial Inclusion, Machine Learning, Alternative Credit Scoring, Algorithmic Bias, Digital Divide, Ethical AI, Global South

JEL: G21, G23, 016, O33, D81, I25, C45, C55

1. Introduction

Artificial intelligence (AI) comprises computer systems that can perform tasks typically requiring human intelligence such as pattern recognition, decision-making, and language understanding. It is rapidly emerging as a critical driver of financial inclusion by enabling access to essential financial services for historically underserved populations. Financial inclusion represents one of finance's greatest challenges, with two billion adults worldwide unable to access loans, insurance, or banking services because they lack traditional credit histories ((World Bank, 2021). Credit invisibility occurs when individuals have no footprint in conventional credit bureaus—no credit cards, mortgages, or installment loans that demonstrate repayment behavior. Even those with "thin files" face exclusion; traditional FICO scores require at least six months of credit activity across multiple accounts (Consumer Financial Protection Bureau, 2015).

The integration of AI allows financial institutions to break through legacy barriers such as high costs, lack of infrastructure, and information asymmetries, improving the reach, efficiency, and personalization of offerings (Akanfe, Bhatt, & Lawong, 2025; Björkegren & Grissen, 2019; Kshetri, 2021; Vuković, Dekpo-Adza, & Matović, 2025). AI leverages machine learning (algorithms that improve automatically through experience), natural language processing (NLP)—technology enabling computers to understand and generate human language—and algorithmic analytics to empower payments, savings, lending, insurance and investments (Mhlanga, 2020).

This paper systematically explores AI applications in each financial need area, highlighting current research, real-world implementations from the Global South, and implications for inclusive development.

2. Methodology

This study employs a critical review of academic and gray literature, combined with a targeted analysis of multiple real-world case studies from the Global South.

The literature review draws on peer-reviewed articles, working papers, and reports from international organizations (e.g., World Bank, CGAP, OECD) to explore the theoretical and empirical landscape of AI in financial inclusion, with a focus on emerging trends, gaps, and controversies.

The qualitative analysis examines multiple purposively selected case studies of AI-driven financial services—including digital payment platforms (e.g., M-Pesa, GCash), micro-lending apps (e.g., Tala, Branch), and insurtech solutions (e.g., BIMA, Pula)—to identify recurring patterns, operational challenges, and humane dilemmas. For this analysis, the BHAI framework is adopted (Arvind Ashta, 2025). The BHAI philosophy advocates for humane AI development through multidimensional inclusion—political, economic, and social—alongside environmental consciousness, calling for adaptive institutional oversight that guides how technologies are applied rather than rigidly controlling innovation. It emphasizes context-sensitive governance with transparent accountability to ensure AI advances benefit all people, especially marginalized populations, while addressing interconnected technological, environmental, and societal challenges through the lens of dignity, ethics, and social impact. The framework operationalizes this vision through six core components—innovation assessment, human dignity and justice, ethical oversight, inclusion and social impact, contextual sensitivity drawing on literary and philosophical sources, and actionable reflection—using an interpretative-constructivist approach that prioritizes context-rich, nuanced insights over universal generalizations to guide humane and equitable AI development (Arvind Ashta, 2025).

By triangulating these sources, the paper offers a nuanced assessment of AI's role in microfinance, while acknowledging the limitations of a non-systematic review approach.

3. Findings

3.1 Payments

3.1.1 Digital Identity Verification & Onboarding

The adoption of AI-powered digital Know Your Customer (KYC) processes in payment platforms has fundamentally transformed onboarding experiences for unbanked populations by automating identity verification using biometric scans (fingerprints, facial recognition, iris scans), document analytics powered by computer vision (AI systems that can 'see' and interpret images), and liveness detection (technology that confirms a person is physically present rather than using a photo or video) (Kuraku, Gollangi, & Sunkara, 2020). These systems typically employ supervised learning—where algorithms are trained on labeled datasets of genuine versus fraudulent documents—combined with deep learning neural networks (interconnected layers of algorithms inspired by the human brain) for facial recognition.

Mobile money providers like M-Pesa[1] in Kenya have pioneered AI-enhanced KYC that allows users to register using basic mobile phones and biometric verification, reaching millions of previously unbanked individuals. In India, the Aadhaar biometric identity system[2], integrated with payment platforms like Paytm and PhonePe, uses AI-powered iris and fingerprint matching to onboard over 400 million digital payment users[3]. NuBank was able to reach millions of unbanked people in Brazil through its online credit card which used automated KYC checks (Chu, Laranjeira, & Levindo, 2020).

[1] https://www.safaricom.co.ke/annualreport_2022/our-technology/
[2] https://www.biometricupdate.com/202508/uidai-celebrates-2b-aadhaar-face-biometric-authentications-milestone
[3] https://www.indiatvnews.com/technology/news/google-pay-phonepe-and-paytm-users-now-use-your-face-or-fingerprint-for-upi-payments-making-them-pin-free-2025-10-08-1011756

These innovations minimize manual errors, dramatically accelerate access to financial services from weeks to minutes, and increase trust in digital payments for those previously excluded from the formal financial sector while ensuring regulatory compliance and scalability even in markets characterized by limited infrastructure.

3.1.2 Fraud Detection & Transaction Security

Machine learning models enable sophisticated real-time fraud detection in payment platforms, identifying anomalies, fraudulent activities, and irregular behavioral patterns as transactions occur across networks (Mhlanga, 2020). These systems predominantly use supervised learning trained on historical fraud cases, combined with unsupervised learning (algorithms that find hidden patterns in data without labeled examples) to detect novel fraud schemes not seen before. Advanced implementations employ ensemble methods (combining multiple AI models for better accuracy) that perform better than a single classifier (Dietterich, 2000). Anomaly detection algorithms can flag deviations from normal transaction patterns.

Flutterwave, a Nigerian payment gateway processing transactions across Africa, uses AI-powered fraud detection that analyzes multiple behavioral and transactional features in real-time, reducing fraud losses while enabling cross-border payments for small businesses[4]. GCash in the Philippines employs machine learning to monitor transactions of over 94 million users[5], detecting suspicious patterns like account takeovers or merchant fraud that would be impossible for human analysts to identify at scale. Kenya's Safaricom uses AI to protect M-Pesa transactions from SIM-swap fraud and account compromise, analyzing device fingerprints, location patterns, and behavioral biometrics[6].

By processing massive datasets with unprecedented speed and accuracy, these AI systems can flag suspicious activity instantly, dynamically adapting risk thresholds and security protocols to evolving user behavior patterns and emerging threat landscapes, which significantly improves payment security while building customer confidence, especially critical for first-time users entering the digital financial ecosystem.

3.1.3 Personalization & User Segmentation

AI enables payment systems to tailor services precisely to individual users through clustering algorithms (unsupervised learning methods that group similar users together) and classification models (supervised learning that predicts user categories) based on transactional patterns, behavioral analytics, geographic data, and temporal usage trends. Recommendation systems, similar to those used by e-commerce platforms, employ collaborative filtering (predicting user preferences based on similar users' behavior) to suggest relevant payment features, cashback offers, and financial products.

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[4] https://thepaymentsassociation.org/article/flutterwave-2024-report-highlights-record-growth-expansion-and-innovation/

[5] https://www.philstar.com/opinion/2024/11/12/2399421/gcash

[6] https://kenyanwallstreet.com/how-safaricom-is-leveraging-ai-to-bolster-m-pesa-security-and-efficiency

WeChat Pay and Alipay in China pioneered AI-driven personalization in digital payments (Torre & Xu, 2020), analyzing billions of transactions to offer customized merchant recommendations, targeted promotions, and differentiated service tiers based on user behavior. India's PhonePe employs AI to customize its interface and product recommendations for diverse user segments, from urban professionals to rural farmers, ensuring relevance across literacy levels and use cases[7].

Platforms can offer customized payment options, personalized loyalty rewards, and differentiated access mechanisms, rising above traditional one-size-fits-all approaches that often fail to meet diverse user needs. This sophisticated personalization builds stronger user engagement, encourages sustained participation in the financial ecosystem, and enables rapid response to evolving customer needs, ensuring that financial services genuinely serve the populations they aim to reach.

3.1.4 Embedded Finance & Cross-Platform Integration

AI-driven embedded payment solutions facilitate seamless integration with other essential daily services including utility management, e-commerce platforms, public transportation systems, or mobile wallet applications, substantially amplifying reach and user convenience (Bélanger, Ashta, & Mason, 2025). These systems use API-based integration (application programming interfaces that allow different software systems to communicate) enhanced with AI-powered risk scoring and real-time fraud detection operating across platforms. Reinforcement learning (algorithms that learn optimal strategies through trial and error with rewards for successful outcomes) can optimize payment routing, fee structures, and approval rates across integrated channels.

Grab in Southeast Asia embeds AI-enhanced payments within ride-hailing, food delivery, and financial services, processing millions of micro-transactions daily across eight countries with intelligent fraud prevention adapted to each use case[8]. Indonesia's Gojek similarly integrates payments across transportation, food, logistics, and financial services using AI to detect fraudulent merchants and protect users across ecosystem touchpoints[9]. In Africa, companies like Jumia and Konga embed AI-powered payment scoring into e-commerce checkout, enabling instant credit decisions for purchases. Jumia is driving its brand with AI-driven customer service and product recommendation, while Konga is deploying predictive analytics and personalised marketing[10].

Real-time fraud detection capabilities and adaptive user authentication processes minimize security risks while enabling service providers to deliver faster, more reliable payment experiences across multiple touchpoints. By embedding financial services within familiar contexts rather than requiring separate banking interactions, AI helps overcome psychological

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[7] https://brandwell.ai/blog/how-to-customize-phonepe-payment-solutions-with-ai-complete-guide/
[8] https://bytebridge.medium.com/comprehensive-report-on-grab-holdings-daa9e7d3918f
[9] https://techwireasia.com/2020/11/gojek-sees-profitability-ahead-after-a-decade-of-rapid-growth/
[10] https://thenationonlineng.net/revealed-nigerian-brands-banking-on-robots-ai/

barriers to adoption and creates natural pathways for previously excluded populations to enter the formal financial system. In one study, 70% reported that the use of artificial intelligence in risk assessment and social impact evaluation improves decision making (Manta, Vasile, & Rusu, 2025).

3.1.5 Humane Considerations

While AI transforms payment accessibility, critical ethical concerns demand attention. Dignity and justice require that biometric data collection respects privacy and obtains informed consent, particularly from vulnerable populations who may not fully understand data implications or have limited alternatives.

Cases of biometric data breaches in India's Aadhaar system and concerns about Chinese payment platforms' data practices highlight risks of surveillance capitalism where user data becomes a commodity (Zuboff, 2019).

Governance frameworks must ensure AI systems do not perpetuate discrimination through biased algorithms that systematically disadvantage certain demographics—for instance, fraud detection models trained predominantly on urban transaction patterns may incorrectly flag legitimate rural transactions as suspicious, effectively excluding rural users (Solon Barocas & Selbst, 2016).

Inclusion efforts must bridge digital divides by ensuring that AI-powered payment solutions remain accessible to those with limited digital literacy, unreliable internet connectivity, or older devices—many AI-enhanced payment apps require smartphones and stable internet, potentially excluding the most marginalized (V. Bumacov, Ashta, & Singh, 2014; Hurley & Adebayo, 2016).

Transparency in algorithmic decision-making, particularly around why accounts are frozen or transactions blocked, robust data protection measures that prevent unauthorized sharing of financial data, and accessible recourse mechanisms when AI systems make errors are essential to building trust. They ensure that AI-driven payment innovations genuinely empower rather than exploit marginalized communities (Akanfe et al., 2025).

3.2 Savings & Pensions

3.2. 1 Behavioral Nudges & Automated Savings

AI's capacity for micro-segmentation and predictive analytics enables savings platforms to create highly personalized savings mechanisms that adapt to individual users' unique employment patterns, income volatility cycles, and financial goals. These systems employ supervised learning models trained on historical savings behavior to predict optimal savings amounts and timing, combined with reinforcement learning algorithms that continuously optimize nudging strategies based on user responses. Behavioral economics principles are operationalized through AI-powered interventions such as automated round-up features (saving spare change from transactions), goal-based savings trackers, and timely motivational messages(Cook & McKay, 2015; Makunda & Matiko, 2023). Table 1 shows some of the behavioral finance features incorporated by digital savings accounts.

Table 1: Behavioural finance nudges in digital savings

Behavioural Principle	Digital Savings Nudge	Result
Present Bias	Makes saving instant & frictionless.	Reduces procrastination.
Mental Accounting	Creates a separate "Savings" account.	Protects savings from daily spending.
Loss Aversion	Offers interest (a "gain").	Frames “not saving” as a loss.
Commitment Device	"Lock Savings" feature.	Enforces discipline by restricting access.
Feedback Loop	Links savings to loan eligibility.	Rewards saving with a tangible, valuable

Now, artificial intelligence can take this further. By analysing the payment patters based on bank of mobile payments, AI can analyze transaction patterns and automatically suggest personalized savings amounts based on predicted future income, enabling millions of informal workers to build emergency funds. Savings and investement fintechs can use machine learning to create customized automated savings plans that adapt to users' irregular income streams.

In India, micro-savings platforms like Jar use AI to invest small amounts automatically in digital gold, with algorithms determining optimal purchase timing based on price patterns and user affordability[11]. These AI-driven interventions motivate users to save regularly by aligning financial products with their actual circumstances rather than imposing rigid structures, improving client loyalty and helping close critical inclusion gaps in retirement planning and emergency savings, ultimately contributing to more financially secure populations across diverse socioeconomic contexts.

3.2.2 Pension Optimization & Retirement Planning

Generative AI and simulation models can forecast personalized retirement outcomes for individual users, creating detailed scenario analyses showing how different savings strategies, contribution levels, and investment allocations affect long-term financial security. These systems use Monte Carlo simulations (running thousands of possible future scenarios with

[11] https://www.forbesindia.com/article/leadership/jar-how-to-build-your-own-pot-of-gold/93180/1

different economic conditions) combined with supervised learning models that predict life expectancy, healthcare costs, and inflation impacts based on individual characteristics. Time-series forecasting algorithms (models specialized in predicting sequential data points) project pension growth under various economic scenarios.

In Latin America, Mexico's CONSAR (pension regulator) has piloted AI tools that help informal workers understand retirement needs through simplified projections based on irregular income patterns. It has also used AI to detect potentially suspicious websites aimed at pension savers [12].

According to a CFA Institute report (Hayman, 2024), AI can be applied across the pension value chain to enhance personalization, efficiency, and accuracy in addressing key retirement system challenges. AI applications include improving member engagement through chatbots and personalized communications, automating recordkeeping and fraud detection, streamlining governance processes through document analysis and covenant assessments, enhancing investment decision-making via predictive analytics and portfolio optimization, and supporting better decumulation strategies in the payout phase. The technology can help pension funds analyze large datasets to improve trustee decision-making, provide personalized retirement planning tools tailored to individual member characteristics, reduce administrative burdens, and enhance both DB and DC plan management through machine learning models that optimize asset allocation and risk management.

South Africa's pension administrators are implementing AI-driven tools to project retirement adequacy for workers transitioning between formal and informal employment. This adaptive guidance significantly enhances financial understanding across literacy levels, increases meaningful engagement with retirement planning, and empowers users to make confident, informed decisions about their long-term financial stability. By translating complex financial concepts into accessible scenarios and visualizations, AI strengthens the inclusive impact of digital savings platforms, making sophisticated financial planning tools available to populations that previously lacked access to professional financial advice.

3.2.3 Humane Considerations

AI-driven savings innovations must uphold human dignity by ensuring that automated nudging systems do not exploit vulnerable savers through manipulative dark patterns (design choices that trick users into actions against their interests) or opaque fee structures disguised as helpful features. Governance mechanisms must prevent algorithmic manipulation that encourages excessive risk-taking or inappropriate long-term lockups for users who may face emergencies requiring liquidity—several African digital savings platforms have faced criticism for making withdrawals difficult while aggressively pushing savings deposits.

Ethical frameworks should ensure that personalization respects cultural values around money, family obligations, and retirement expectations rather than imposing Western financial planning

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[12] https://www.nortonrosefulbright.com/en-419/knowledge/publications/ba2b4dbd/pensions-regulator-develops-ai-tool-to-detect-potentially-suspicious-pension-websites

models—in many cultures, supporting extended family takes precedence over individual retirement savings, yet AI systems may penalize such transfers as 'poor savings discipline' (Anderson, Baland, & Moene, 2009).

Inclusion requires addressing the reality that informal workers often face irregular income streams, making rigid automated savings schedules counterproductive—AI systems should accommodate rather than penalize income variability.

The CFA report (Hayman, 2024) emphasizes that while AI offers significant potential to improve retirement outcomes and operational efficiency, successful implementation requires balancing automation with human oversight, ensuring data privacy and security, maintaining transparency and explainability in AI-driven decisions, and using technology to augment rather than replace human judgment in fiduciary responsibilities. Transparent communication about investment performance, accessible withdrawal options during genuine emergencies without excessive penalties, protection against algorithmic errors that could devastate retirement security (such as incorrect contribution calculations or failed transfers), and ensuring that predictive models don't discriminate based on factors like gender or geography when projecting retirement needs are fundamental requirements for ethical AI deployment in savings and pensions.

3.3 Lending & Credit

3.3.1 Alternative Credit Scoring & Data Analytics

A credit score is a number based on an analysis of a person's credit files, to represent the creditworthiness of an individual. It has been used in developed countries for many decades but was introduced more recently to microcredit (Vitalie Bumacov, Ashta, & Singh, 2017). Credit invisibility occurs when individuals have no footprint in conventional credit bureaus: no credit cards, mortgages, or installment loans that demonstrate repayment behavior. Traditional credit scoring creates a paradox that locks out entire populations: you cannot get your first loan without credit history, but you cannot build credit history without getting loans (World Bank, 2021).

In the United States alone, 45 million adults are either completely credit invisible or have credit files too thin to generate reliable scores, representing 20% of the adult population, disproportionately concentrated among minority communities and lower-income households (Consumer Financial Protection Bureau, 2015).

Now, AI leverages alternative data sources (see Table 2) including mobile phone usage patterns, e-commerce transaction histories, utility payment records, social network data, and smartphone sensor data (like GPS patterns indicating stable employment) for sophisticated credit scoring that addresses exclusion of populations with thin files or those without formal credit histories (Björkegren & Grissen, 2019; Gambacorta, Huang, Qiu, & Wang, 2019; Nuka & Ogunola, 2024).

Research demonstrates that telecommunications payment data shows particularly strong predictive power—studies across multiple markets find correlations of 0.65 to 0.72 between mobile phone bill payment consistency and loan repayment rates, comparable to traditional FICO scores (Björkegren & Grissen, 2019).

Table 2: Traditional versus Alternative Data

Data Type	**Traditional Data**	**Alternative Data**
Income	Formal employment income	Gig economy and freelance earnings
Payment History	Credit card and loan repayments	Rental payments, utility bills
Spending Patterns	Bank account transactions	Mobile money and digital wallets
Assets	Property ownership	Informal savings groups and investments
Demographics	Age, marital status	Social media and online activity
Credit History	Bank credit reports	Alternative lending platform histories
Behavioral Data	N/A	Psychometric testing, online behavior analytics

Source: The first five rows (Nuka & Ogunola, 2024) and the last two rows added by author.

These systems employ supervised learning with classification algorithms such as gradient boosting machines, random forests (ensemble methods that combine multiple decision trees), and neural networks trained on historical repayment data to predict default probability (Jonnalagadda & Babu, 2025). Alternative data scoring overwhelmingly relies on gradient boosted tree methods, particularly XGBoost and LightGBM, which power 70-80% of production systems at companies like Affirm, Upstart, and Kabbage because they automatically detect non-linear relationships and handle the messy, incomplete nature of alternative data (Khandani, Kim, & Lo, 2010).

Where traditional scoring uses 5-10 variables from credit bureaus, ML models might incorporate 500-5,000 features extracted from alternative sources and this reduces default rates (Berg, Burg, Gombović, & Puri, 2019). Feature engineering (the process of selecting and transforming raw data into useful predictors) identifies hundreds of behavioral indicators from alternative data—call patterns, SMS metadata, app usage, mobile money transactions—that correlate with creditworthiness (Gambacorta et al., 2019).

Kenya's Branch and Tala pioneered smartphone-based credit scoring analyzing over 10,000 data points from users' phones to predict repayment likelihood, disbursing billions in microloans to borrowers with no formal credit history. In India, companies like ZestMoney and KreditBee use AI models incorporating e-commerce behavior, digital wallet usage, and education data to score young borrowers entering credit markets. Nigeria's Carbon (formerly Paylater) analyzes bank transaction data, mobile money flows, and social media presence to assess creditworthiness for millions of borrowers (Olajide et al., 2025). China's Ant Financial developed Zhima Credit, which scores over 1 billion users based on consumption behavior, money transfer networks, and fulfillment of commitments[13].

By recognizing reliability demonstrated through non-traditional channels, these AI-powered alternative credit scoring systems fundamentally transform credit accessibility. They open lending markets to billions previously deemed unscoreable or too risky based solely on absence from traditional credit bureaus. This enables entrepreneurs, smallholder farmers, and gig economy workers to access productive capital (Djeundje, Crook, Calabrese, & Hamid, 2021; Jagtiani & Lemieux, 2019; Nuka & Ogunola, 2024).

3.3.2 Automated Credit Risk Assessment

Automated underwriting systems leverage ensemble machine learning methods combining multiple supervised learning algorithms (logistic regression for baseline probability, gradient boosting for complex patterns, and neural networks for non-linear relationships) for fast, accurate risk evaluations in microfinance and small-to-medium enterprise lending (Milana & Ashta, 2021). These systems process loan applications in real-time, analyzing hundreds of variables simultaneously including alternative credit scores, cash flow patterns, business sector risks, and macroeconomic indicators to generate risk ratings and recommend loan terms. Natural language processing analyzes loan applications and business descriptions to assess viability, while computer vision can evaluate collateral photos submitted via mobile apps.

India's Aye Finance uses AI underwriting to assess micro-enterprises' creditworthiness by analyzing GST filings, bank statements, and business characteristics, approving loans within 72 hours for businesses traditional banks reject[14]. Kenya's Musoni uses machine learning models to underwrite agricultural loans based on mobile money transaction history, farm size data from satellite imagery, and weather patterns, serving smallholder farmers efficiently. Brazil's Creditas employs AI to underwrite asset-backed loans for lower-income borrowers using alternative data combined with vehicle and property valuations. Bangladesh's bKash integrates AI-powered micro-lending directly into its mobile money platform, using transaction history to automatically pre-approve small loans for merchants and users.

[13] https://en.wikipedia.org/wiki/Zhima_Credit
[14] https://www.ayefin.com/wp-content/uploads/2024/12/Aye-Finance-Limited-Industry-Report.pdf

By reducing decision-making costs from hundreds of dollars per loan to pennies, and processing times from weeks to minutes, AI-powered underwriting makes credit accessible to borrowers seeking smaller loan amounts that would be unprofitable under traditional manual underwriting processes, enabling financial inclusion at unprecedented scale (Jagtiani & Lemieux, 2019).

3.3.3 Flexible Loan Products & Dynamic Pricing

AI enables lenders to create highly tailored lending products for diverse populations including gig workers, self-employed individuals, seasonal laborers, and small entrepreneurs by using reinforcement learning to continuously optimize loan terms, repayment schedules, and interest rates based on borrower behavior and repayment success. Predictive models employing time-series analysis forecast income volatility and optimal repayment timing for borrowers with irregular cash flows. Dynamic pricing algorithms adjust interest rates based on individual risk profiles, competitive market conditions, and the lender's liquidity needs, making credit more accessible while maintaining profitability.

Argentina's Ualá uses AI to offer flexible credit lines that adapt to users' spending and repayment patterns, with personalized limits and interest rates reflecting individual behavior rather than rigid categorical rules[15]. In Kenya, Apollo Agriculture combines AI credit scoring with flexible repayment schedules aligned to harvest cycles, providing smallholder farmers with inputs financing that accommodates seasonal income patterns[16]. Philippines-based Robocash uses machine learning to dynamically adjust loan terms and pricing based on borrower engagement and partial repayment behavior, reducing defaults while expanding access[17]. India's KreditBee employs AI to offer flexible tenure options and personalized interest rates for young professionals with variable income streams[18].

This sophisticated segmentation moves beyond one-size-fits-all credit products that often fail to serve non-traditional borrowers effectively, recognizing that different economic activities require different financial structures. Personalized lending increases repayment success rates while expanding access to populations whose income variability previously disqualified them from credit, ultimately fostering entrepreneurship and economic development across diverse economic sectors.

3.3.4 Humane Considerations

AI-powered lending raises profound ethical concerns around dignity and justice, particularly regarding algorithmic bias that may perpetuate historical discrimination in credit access (Djeundje et al., 2021). Alternative data can encode existing societal inequalities: low-income

[15] https://www.bnamericas.com/en/news/argentinas-uala-working-on-integrating-2-banks-into-its-tech-stack
[16] https://nation.africa/kenya/news/gender/meet-the-don-teaching-machines-to-speak-africa-s-languages-one-algorithm-at-a-time-5186224
[17] https://juicyscore.ai/en/case-studies/robocash-reduces-high-risk-application-flow-by-75-with-juicyscore
[18] https://www.kreditbee.in/flexi-personal-loan

neighborhoods generate fewer digital footprints simply because residents have older phones, less reliable internet, and lower e-commerce participation.

When machine learning trains on this data, it may learn that sparse digital activity predicts default—not because sparse activity causes default, but because it correlates with poverty. This creates proxy discrimination: the model never sees race, but zip code, device type, and app usage patterns serve as proxies (Solon Barocas & Selbst, 2016). Studies have revealed that even alternative credit scoring models can encode proxy discrimination—for instance, smartphone models, app usage patterns, or social media behavior may correlate with protected characteristics like race or ethnicity, leading to discriminatory outcomes despite not explicitly using these variables.

Governance frameworks must ensure transparency in credit scoring algorithms, including model explainability (the ability to understand why an AI system made a specific decision), allowing borrowers to understand and effectively contest automated decisions affecting their economic opportunities (Bracke, Datta, Jung, & Sen, 2019). Individuals should be granted an opportunity to challenge adverse decisions based on artificial intelligence generated scores (Citron & Pasquale, 2014).

Predatory lending practices can be amplified by AI systems that identify vulnerable populations and target them with exploitative terms disguised as personalized offers. Debt creates an entire industry profiting from exploitation (Rona-Tas & Guseva, 2018).

Inclusion requires that alternative data usage respects privacy boundaries—analyzing call logs, SMS content, or social networks raises serious privacy concerns, especially when users aren't fully informed about data usage (Citron & Pasquale, 2014).

Fair lending principles would require that AI systems do not punish poverty by charging exponentially higher rates to those already economically marginalized: differential pricing should reflect actual risk, not merely exploit price insensitivity among desperate borrowers.

Algorithmic credit expansion must not create debt traps through inappropriate loan approvals or inflexible collection practices, but rather provide genuine pathways to economic empowerment through responsible, affordable credit that borrowers can realistically repay (A. Ashta & Hudon, 2012).

3.4 Insurance

3.4.1 Risk Profiling & Insurability Assessment

AI builds comprehensive insurability profiles by assessing digital footprints and alternative data to evaluate risk levels when traditional actuarial data is limited or nonexistent, enabling insurers to expand coverage into previously underserved markets (Vuković et al., 2025). These systems employ supervised learning with classification models (typically ensemble methods like XGBoost or neural networks) trained on claims history, lifestyle indicators extracted from mobile usage, geographic risk factors from satellite imagery, and health proxies derived from activity patterns. Computer vision analyzes submitted photos of homes, vehicles, or farms to assess insurability and detect pre-existing damage, while satellite imagery and weather data inform agricultural insurance risk models (Lobell, Thau, Seifert, Engle, & Little, 2015).

BIMA, operating across Africa and Asia, uses AI-powered micro-insurance assessment based on mobile phone data and basic health questions to provide life and health insurance to over 45 million low-income customers who lack traditional medical records[19]. In India, Toffee Insurance uses machine learning to offer bite-sized insurance products personalized to individual risk profiles derived from smartphone data and payment histories[20]. Kenya's M-Pesa partners with insurance providers to offer micro-insurance scored based on transaction patterns and mobile usage, automatically enrolling users in appropriate coverage tiers.

By recognizing patterns that correlate with insurance claims experience across diverse data sources, AI enables evidence-based risk pricing that makes insurance economically viable for both providers and low-income customers. This innovation transforms insurance accessibility by moving beyond stereotypes and limited data to data-driven individual risk assessment, opening insurance markets to billions previously deemed too risky or expensive to serve profitably.

3.4.2 Automated Underwriting & Claims Processing

Machine learning automates complex insurance underwriting and accelerates claims processing through computer vision for damage assessment, natural language processing for claims documentation analysis, and predictive models for fraud detection. Supervised learning models trained on historical claims data classify claim validity, estimate loss amounts, and detect fraudulent patterns. Deep learning with convolutional neural networks (specialized architectures for analyzing images) can assess crop damage, vehicle accidents, or property destruction from photos submitted via mobile apps, providing near-instant loss estimates. In parametric insurance (coverage that pays out automatically when specific measurable events occur), AI monitors trigger conditions like rainfall levels, earthquake magnitude, or temperature thresholds using IoT sensors and satellite data.

In Nigeria Pula uses AI-powered parametric insurance for smallholder farmers, automatically triggering payouts based on satellite-derived vegetation health indices and weather station data, eliminating lengthy manual claims processes (Hernandez, Goslinga, & Wang, 2018). Kenya's Britam Insurance employs computer vision to assess vehicle damage from smartphone photos, reducing claims processing time from weeks to hours[21]. India's Acko uses fully

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[19] https://www.cgap.org/about/people/bima
[20] https://wishboxstudio.in/toffee-insurance-a-case-study/
[21] https://thedailywhistle.co.ke/faster-claims-fewer-headaches-ai-is-transforming-kenyas-insurance-industry/

automated claims processing with AI analyzing photos, policy terms, and repair cost databases to approve and disburse claims within minutes for cyber insurance and product protection[22].

By processing risk factors continuously and updating models with emerging data, AI enables dynamic pricing that reflects actual risk rather than outdated assumptions, making innovative insurance products financially sustainable and allowing insurers to offer coverage for climate-related disasters, crop failures, and health emergencies that disproportionately affect poor communities.

3.4.3 Micro-Insurance Products & Parametric Coverage

AI creates highly flexible micro-insurance products and specialized offerings like harvest-linked agricultural insurance, weather-indexed coverage, and health micro-insurance tailored for farmers, informal workers, and micro-entrepreneurs facing context-specific risks. Machine learning models employing regression analysis (predicting continuous numerical outcomes) and time-series forecasting assess correlations between observable parameters (rainfall, temperature, vegetation indices) and actual losses, enabling parametric triggers that are objective and verifiable (Burke & Lobell, 2017). Clustering algorithms segment users based on occupation, location, and risk exposure to design targeted products.

Nigeria's Pula Advisors partners with mobile money platforms to offer index-based crop insurance integrated directly into agricultural input purchases, with AI determining coverage amounts and monitoring weather patterns to trigger automatic payouts via mobile money without requiring manual claims (Hernandez et al., 2018). In Ghana, ACRE Africa uses machine learning combined with satellite data and weather station networks to offer parametric insurance covering drought and excess rainfall risks for smallholder farmers, with payouts automatically triggered when rainfall deviates from optimal levels (Waithaka, Kramer, Kivuva, & Cecchi, 2023). India's Skymet combines AI weather prediction models with blockchain-based smart contracts to automate parametric insurance payouts for millions of farmers, reducing administrative costs and preventing payout delays[23]. Kenya's Turaco offers device insurance, life insurance, and health coverage bundled with mobile money services, using AI to price policies dynamically based on risk profiles[24].

This customization increases insurance value for policyholders while improving risk pools for insurers, creating sustainable insurance markets that protect vulnerable populations from shocks that could otherwise destroy livelihoods.

3.4.4 Humane Considerations

AI-driven insurance must respect dignity by avoiding discriminatory risk profiling that unfairly penalizes vulnerable populations for factors beyond their control. Algorithmic redlining concerns are particularly acute in insurance—AI systems trained on historical data may systematically

[22] https://digiqt.com/blog/acko-insurance-automation/
[23] https://www.skymetweather.com/corporate/cropinsurance.html
[24] https://www.turaco.insure/about-us

deny coverage or charge prohibitive premiums to marginalized communities, perpetuating rather than addressing inequality (Solon Barocas & Selbst, 2016). For instance, using geolocation data may result in higher premiums for residents of informal settlements regardless of individual risk factors, effectively excluding the poor. Governance mechanisms must prevent proxy discrimination where AI uses variables like smartphone type, social network characteristics, or consumption patterns that correlate with protected characteristics like race, religion, or caste.

Justice would require that micro-insurance pricing remains genuinely affordable and that claims processing does not become opaque or systematically deny legitimate claims through algorithmic gatekeeping—some parametric insurance schemes have faced criticism for setting triggers that rarely activate despite farmers experiencing losses. Commitment to expanding genuine protection rather than simply extracting premiums from poor communities requires that insurance products actually transfer meaningful risk and provide timely payouts when losses occur, not just collect premiums while minimizing payouts through restrictive AI-determined conditions.

Inclusion requires recognizing that alternative data usage for risk assessment must not become invasive surveillance that commodifies poverty or exploits vulnerable populations by extracting intimate behavioral data in exchange for basic insurance coverage (Zuboff, 2019).

Ethical deployment demands transparency about how AI determines insurability, with clear explanations of why someone might be denied coverage or charged higher premiums (S. Barocas, Hardt, & Narayanan, 2019). Accessible appeals processes for denied claims or disputed risk ratings are essential, ensuring that algorithmic decisions can be challenged.

3.5 Investments

3.5.1 Automated Portfolio Management & Robo-Advisory

AI-powered robo-advisors (automated platforms providing financial planning and investment management) democratize access to sophisticated portfolio management previously available only through expensive human advisors (Kshetri, 2021). These systems employ supervised learning algorithms for asset allocation (determining optimal distribution across stocks, bonds, and other investments), regression models for return prediction, and optimization algorithms (mathematical methods for finding the best solution under constraints) to build diversified portfolios aligned with individual risk tolerance and goals. Reinforcement learning can continuously adjust strategies based on market conditions and portfolio performance, learning optimal rebalancing policies over time. Natural language processing enables conversational interfaces that guide novice investors through onboarding and ongoing portfolio management in accessible language.

Brazil's Nubank offers AI-driven investment products to over 120 million customers in Brazil, Mexico, and Colombia, many first-time investors, with automated portfolio construction based on individual risk profiles and goals[25]. Mexico's Flink uses robo-advisory with micro-investing features, allowing users to start with as little as 30 pesos while AI builds diversified portfolios adapted to Latin American markets[26]. Kenya's Ndovu provides automated investment management accessible via mobile money integration, using AI to create portfolios suitable for African market conditions and investor profiles ranging from urban professionals to rural savers[27]. India's Groww platform use machine learning to recommend thematic investment portfolios and mutual funds tailored to users' financial situations, education levels, and investment horizons[28].

By providing professional-grade portfolio management at fraction of traditional costs, AI enables small-scale investors to participate in wealth-building opportunities historically reserved for the affluent.

3.5.2 Risk-Based Asset Allocation & Market Intelligence

Machine learning systems comprehensively assess investor risk profiles and dynamically allocate assets using portfolio optimization algorithms that balance expected returns against volatility. These systems employ supervised learning for risk tolerance classification, time-series forecasting models (LSTM neural networks—long short-term memory networks specialized for sequential data—and ARIMA models) to predict market movements (Krishnan, Ashta, & Babu, 2021), and reinforcement learning to discover optimal long-term investment strategies that adapt to changing market regimes. Sentiment analysis using natural language processing extracts market signals from news articles, social media, and financial reports to inform investment decisions. Volatility prediction models and correlation analysis ensure diversification across assets with different risk characteristics.

Colombia's Tyba uses AI to assess users' risk profiles through behavioral questions and financial data, automatically allocating investments across diverse assets while continuously rebalancing based on market conditions and individual circumstances[29]. Chile's Fintual employs machine learning to construct portfolios optimized for different life stages and goals, with AI-driven rebalancing responding to market volatility while maintaining target risk levels[30]. In India, ET Money uses AI to recommend mutual fund portfolios combining risk assessment with tax optimization strategies, while continuously monitoring portfolio health and suggesting

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[25] https://international.nubank.com.br/company/with-122-million-customers-nubank-creates-products-capable-of-gaining-global-scale/
[26] https://www.latamfintech.co/articles/mexican-neobroker-flink-raised-57-m-in-a-series-b-round-to-boost-financial-inclusion-in-latam
[27] https://www.ndovu.co/about-ndovu-rw
[28] https://yourstory.com/2019/11/groww-leverages-technology-eliminate-hassle
[29] https://alpaca.markets/blog/tyba-creating-investment-access-in-latin-america/
[30] https://www.hi.vc/insights/an-ai-tool-for-analyzing-investment-statements-fintuals-new-bet

adjustments[31]. Nigeria's Risevest allows diaspora and local investors to access global markets through AI-curated portfolios that manage currency risk and optimize returns for naira-based investors[32].

For novice investors lacking sophisticated financial knowledge, automated risk management provides professional-grade portfolio oversight, helping them build wealth over time while avoiding common pitfalls like panic selling during downturns, excessive concentration in high-risk assets, or overly conservative allocations that fail to generate real returns after inflation.

3.5.3 Humane Considerations

AI-powered investment platforms must uphold dignity by ensuring that democratized access does not expose financially vulnerable populations to inappropriate risks through algorithmic recommendations that prioritize platform profits over investor welfare. Governance frameworks must prevent conflicts of interest where AI systems recommend high-fee products or investments generating higher commissions for platforms rather than better returns for investors—some robo-advisors have faced criticism for steering users toward proprietary funds with higher fees. Ethical deployment demands that AI investment tools genuinely serve wealth-building goals rather than extracting fees from novice investors through frequent trading or complex products (Kalluri, 2020). Clear disclosure of conflicts of interest, commitment to fiduciary responsibility (legal obligation to act in clients' best interests) in automated investment advice, and protection against algorithmic failures that could cause significant portfolio losses are essential.

Justice requires transparency in how algorithms allocate investments and assess suitability, with clear explanations of automated decisions affecting people's financial futures. Model explainability is crucial—investors should understand why they received specific recommendations and what assumptions drive their portfolio allocation (Bracke et al., 2019).

Inclusion requires recognizing that investment literacy varies dramatically, necessitating protective guardrails against algorithmic manipulation while preserving investor autonomy—overly aggressive risk profiling might push unsophisticated investors into volatile assets they don't understand, while overly conservative approaches might leave them unable to build wealth.

Ensuring that robo-advisors account for local market conditions, currency risks, and tax implications relevant to Global South investors—rather than simply adapting algorithms designed for developed markets—is crucial for genuine financial inclusion in investment services.

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[31] https://www.etmoney.com/tax-saving

[32] https://risevest.com/why-rise

4. Discussion

This systematic examination of AI applications across five critical financial sectors reveals both remarkable technological advances and persistent ethical challenges that must be addressed to ensure truly inclusive and humane financial systems. The evidence demonstrates that AI is not merely automating existing processes but fundamentally transforming how financial services can reach and serve marginalized populations.

4.1 AI Technologies Across Financial Sectors

To understand the diverse AI applications described in this paper's findings, it is essential to first grasp the foundational machine learning paradigms that underpin these systems. Machine learning—the core technology driving AI-enabled financial inclusion—comprises three primary learning approaches, each suited to different types of problems:

Supervised learning involves training algorithms on labeled datasets where the correct answer is known (for example, historical loan data labeled as "repaid" or "defaulted"). The algorithm learns patterns that map inputs to outputs, enabling it to make predictions on new, unseen data. This approach powers most classification tasks (is this transaction fraudulent?) and regression tasks (what return can be expected?).

Unsupervised learning discovers hidden patterns in data without pre-labeled examples. Rather than predicting a known outcome, these algorithms identify natural groupings, detect anomalies, or find underlying structures in data. This approach excels at segmentation (grouping similar customers) and anomaly detection (finding unusual patterns that might indicate fraud).

Reinforcement learning takes a fundamentally different approach: algorithms learn optimal strategies through trial and error, receiving rewards for successful actions and penalties for unsuccessful ones. Over many iterations, the system discovers which decisions lead to the best long-term outcomes. This paradigm is particularly powerful for sequential decision-making problems like portfolio management or dynamic pricing.

Within each of these three paradigms, specific techniques and algorithms have proven particularly effective for financial inclusion applications. Additionally, certain cross-cutting technologies—natural language processing and computer vision—can employ any of these learning paradigms depending on the specific task. Table 3 organizes all AI technologies discussed in this paper's findings section according to these learning paradigms and shows which techniques are deployed across the five financial sectors.

Table 3: AI Technologies by Financial Sector (Hierarchical Structure)

Learning Paradigm	Specific Technique/Application	Payments	Savings & Pensions	Lending & Credit	Insurance	Investments

SUPERVISED LEARNING	Classification (fraud, risk, default)	KYC verification, fraud detection	Savings behavior prediction	Credit scoring, default prediction	Risk classification, claims validity	Risk tolerance classification
	Gradient Boosting (XGBoost/LightGBM)	-	-	Alternative credit scoring (70-80% of systems)	Risk profiling	-
	Neural Networks (CNNs for images)	Biometric recognition (face, iris, fingerprint)	-	-	Damage assessment from photos	-
	Regression Models	-	-	-	Parametric loss prediction	Return prediction
	Time-Series Forecasting (labeled)	-	Pension projections (ARIMA)	Income volatility prediction	-	Market prediction (LSTM, ARIMA)
UNSUPERVISED LEARNING	Anomaly Detection	Novel fraud pattern detection	-	Novel fraud schemes	-	-
	Clustering/Segmentation	User segmentation, personalization	-	Borrower segmentation	Risk-based groups	Investor profiling
REINFORCEMENT LEARNING	Strategy Optimization	Payment routing, fee optimization	Nudging strategy optimization	Dynamic pricing	-	Portfolio strategy optimization
CROSS-CUTTING TECHNIQUES	Natural Language Processing	-	Chatbots for guidance	Application analysis	Claims documentation	Sentiment analysis, chatbots
	Computer Vision	Document verification, liveness detection	-	Collateral photos	Satellite imagery, damage photos	-
STATISTICAL METHODS	Monte Carlo Simulation	-	Retirement scenarios	-	-	Risk scenarios

This hierarchical organization reveals several critical patterns about how AI technologies are being deployed for financial inclusion:

First, supervised learning dominates across all five sectors, reflecting both its technical maturity and the availability of labeled training data in financial services. Every sector employs supervised classification or regression for core predictive tasks—fraud detection in payments, credit scoring in lending, risk assessment in insurance, and return prediction in investments. This prevalence indicates that financial institutions possess sufficient historical data (transactions, defaults, claims, returns) to train supervised models effectively. The success of supervised learning in financial inclusion stems from its ability to learn from past patterns and apply those lessons to new cases at scale.

Second, gradient boosting methods—specifically XGBoost and LightGBM—have emerged as the dominant technique for alternative credit scoring, powering 70-80% of production systems in lending as documented in the findings. This near-universal adoption is not accidental. These ensemble methods excel at handling the messy, incomplete, heterogeneous nature of alternative data (mobile phone metadata, e-commerce transactions, utility payments) that characterizes credit-invisible populations. Unlike simpler models that assume clean, structured data, gradient boosting automatically detects complex non-linear relationships, handles missing values gracefully, and combines insights from multiple decision trees to achieve robust predictions. This technical advantage explains why virtually every successful alternative lending platform—from Kenya's Tala and Branch to India's ZestMoney and KreditBee—relies on gradient boosting as its core technology.

Third, deep learning neural networks find specific applications where they provide unique capabilities, particularly in payments and insurance. Convolutional Neural Networks (CNNs) power biometric authentication systems that enable digital identity verification for unbanked populations—analyzing facial features, iris patterns, and fingerprints to confirm identity without requiring traditional documentation. Similarly, CNNs assess damage from photographs in insurance claims processing, extracting insights from visual data that traditional algorithms cannot process. The pattern is clear: deep learning is deployed where the input data is fundamentally unstructured (images, audio, video) rather than tabular, and where its superior pattern recognition capabilities justify the higher computational costs and data requirements.

Fourth, unsupervised learning plays a more limited but critical role, primarily in fraud detection and customer segmentation. Anomaly detection algorithms identify novel fraud patterns that supervised models—trained only on historical fraud examples—would miss entirely. This capability is essential because fraudsters constantly adapt their tactics; purely supervised approaches would always lag behind. Similarly, clustering algorithms enable customer segmentation without requiring pre-defined categories, allowing platforms to discover natural groupings in their user bases and personalize services accordingly. The combination of supervised learning (for known patterns) and unsupervised learning (for unknown patterns) provides more comprehensive coverage than either approach alone.

Fifth, reinforcement learning remains relatively rare, appearing primarily in optimization contexts—payment routing, automated savings nudging, dynamic loan pricing, and portfolio management. This limited deployment reflects reinforcement learning's significant challenges: it requires extensive trial-and-error learning (potentially costly if errors harm real customers), demands careful reward function design (mis-specified rewards can lead to perverse outcomes), and exhibits sample inefficiency (requiring many iterations to learn effective policies). Where reinforcement learning does appear, it addresses sequential decision-making problems where the optimal action depends on long-term consequences rather than immediate predictions, and where simulation or safe experimentation is possible.

Sixth, cross-cutting technologies—Natural Language Processing and Computer Vision—bridge multiple learning paradigms and extend AI's reach into previously inaccessible data modalities. NLP enables conversational interfaces (chatbots providing pension guidance or investment advice), analyzes unstructured text (loan applications, business descriptions, claims documentation), and extracts market sentiment from news and social media. Computer Vision processes identity documents, assesses collateral from photos, evaluates crop damage from satellite imagery, and verifies physical presence through liveness detection. These technologies transform unstructured human-generated content into structured data that supervised, unsupervised, or reinforcement learning algorithms can then process.

Finally, the table reveals important sector-specific technological profiles. Payments emphasizes real-time supervised classification (fraud detection) combined with biometric neural networks and reinforcement learning for routing optimization—reflecting the sector's need for instant decisions and continuous system optimization. Lending concentrates supervised learning intensity, particularly gradient boosting for alternative credit scoring, reflecting the core challenge of predicting creditworthiness from diverse data sources. Insurance uniquely combines supervised classification with computer vision for damage assessment and regression models for parametric triggers, reflecting its need to process both visual evidence and quantifiable parameters. Investments employs the most diverse toolkit (supervised classification, time-series forecasting, reinforcement learning, and NLP) reflecting the multi-faceted challenges of asset allocation, market prediction, risk management, and investor communication. Savings & Pensions occupies a middle ground, using supervised learning for behavioral prediction and reinforcement learning for nudging optimization, but lacking the visual or textual analysis needs that drive other sectors.

This systematic mapping reveals that AI-driven financial inclusion is not monolithic but rather comprises a sophisticated ecosystem of complementary technologies, each addressing specific challenges through appropriate learning paradigms. The dominance of supervised learning reflects pragmatic choices based on data availability and problem structure, while the selective deployment of unsupervised learning, reinforcement learning, and advanced techniques like deep learning indicates thoughtful matching of tools to tasks. Understanding this technological landscape is essential for evaluating both the opportunities and risks that AI presents for financial inclusion, as different techniques carry distinct implications for fairness, transparency, and accountability: issues we turn to next in examining humane considerations.

4.2 Humane Challenges Across Financial Sectors

While AI's technical capabilities are impressive, Table 4 reveals that each sector confronts profound ethical challenges that threaten to undermine financial inclusion goals if not adequately addressed.

Table 4: Humane Considerations by Financial Sector

Humane Challenge	Payments	Savings & Pensions	Lending & Credit	Insurance	Investments
Algorithmic Bias & Discrimination	Fraud models flag rural transactions as suspicious	Gender/geography discrimination in projections	Proxy discrimination through alternative data, historical bias perpetuation	Discriminatory risk profiling, algorithmic redlining	Biased risk assessment
Privacy Violations	Biometric data breaches, surveillance capitalism	Data privacy and security concerns	Invasive alternative data collection (call logs, SMS, social networks)	Invasive behavioral surveillance	
Digital Divide & Exclusion	Smartphone/internet requirements exclude marginalized groups	Irregular income penalized by rigid systems	Sparse digital footprints from poverty misread as risk	Geolocation-based premium increases	Limited access for low digital literacy populations
Lack of Transparency	Opaque account freezing/transaction blocking	Opaque fee structures, withdrawal difficulties	Non-explainable credit decisions, black-box scoring	Unclear denial reasons, opaque claims processing	Conflicts of interest, algorithm opacity
Manipulative Practices		Dark patterns, excessive risk encouragement	Predatory lending targeting vulnerable populations	Restrictive parametric triggers, premium extraction	Fee optimization over returns, inappropriate risk pushing
Cultural Insensitivity		Western financial models imposed, family obligations penalized			One-size-fits-all approaches
Inadequate Recourse	No contestation mechanisms for automated decisions	Liquidity lockups during emergencies	No ability to challenge adverse decisions	Denied claims without explanation	Algorithmic failures without accountability
Governance Gaps	Insufficient data protection, weak oversight	Insufficient human oversight in fiduciary decisions	Insufficient fairness auditing, weak explainability requirements	Inadequate disparate impact testing	Weak fiduciary enforcement

The systematic analysis reveals recurring themes across sectors. **Algorithmic bias and proxy discrimination** emerge as perhaps the most pervasive challenge. Even when protected characteristics like race, ethnicity, or gender are explicitly excluded from models, AI systems learn to use proxy variables—zip code, device type, app usage patterns, social network characteristics—that correlate with these protected attributes, thereby perpetuating historical discrimination under the guise of objective assessment. This is particularly pernicious because it operates invisibly: the model never explicitly considers race, yet systematically disadvantages racial minorities.

Privacy violations constitute another cross-cutting concern. The alternative data powering AI-driven financial inclusion—mobile phone metadata, social media activity, e-commerce behavior, location patterns—represents intimate behavioral surveillance that most users neither fully understand nor meaningfully consent to. The Aadhaar biometric breaches in India and concerns about Chinese payment platform data practices illustrate how financial inclusion can become a pathway to surveillance capitalism, where user data becomes a commodity extracted in exchange for financial access.

The digital divide paradoxically threatens to make AI-driven financial inclusion exclusionary. Many AI-enhanced services require smartphones, stable internet connectivity, and digital literacy—prerequisites that exclude the most marginalized populations who need financial inclusion most urgently. When fraud detection systems are trained predominantly on urban digital transaction patterns, they may incorrectly flag legitimate rural transactions as suspicious, effectively excluding rural users. Similarly, sparse digital footprints generated by poverty—older phones, limited internet, low e-commerce participation—get misread by AI as risk indicators, creating a vicious cycle where poverty itself becomes a disqualifying factor.

Lack of transparency and explainability undermines user agency and prevents meaningful contestation of automated decisions. When an AI system denies credit, freezes an account, or increases insurance premiums, users rarely receive comprehensible explanations of why the decision was made or how they might improve their situation. This opacity violates principles of due process and prevents users from effectively challenging erroneous automated decisions that can have devastating consequences for their economic opportunities.

Manipulative practices disguised as personalization represent a particularly insidious risk. AI systems can identify vulnerable populations and target them with exploitative terms—predatory loans at usurious rates, insurance products with restrictive triggers unlikely to activate, investment products generating high fees with poor returns—all optimized through machine learning to maximize provider profits while minimizing payouts. Behavioral nudging can cross the line into manipulation when it encourages excessive risk-taking by vulnerable savers or locks users into illiquid products they cannot access during emergencies.

Cultural insensitivity reflects how AI models trained predominantly on Western financial behaviors may misinterpret culturally appropriate patterns as risk indicators. In many cultures, supporting extended family takes precedence over individual savings, yet AI systems may penalize such transfers as "poor financial discipline." Irregular income reflecting seasonal agriculture gets flagged as "income instability" rather than recognized as normal for smallholder farmers. These cultural blind spots can make ostensibly inclusive AI systems systematically disadvantage non-Western populations.

4.3 The Inclusion Paradox

Perhaps the most troubling finding is what might be termed the "inclusion paradox": AI enables access to financial services for previously excluded populations, but often at exploitative terms. Alternative credit scoring allows lending to the credit invisible, but frequently at 30-40% APR or higher—rates that provide access while potentially trapping borrowers in debt spirals. Micro-insurance reaches poor populations, but with parametric triggers set to rarely activate despite farmers experiencing losses. Automated investment platforms democratize wealth management, but may steer novice investors into high-fee proprietary products.

This paradox raises fundamental questions about whether AI-driven expansion of financial access should be celebrated as inclusion or critiqued as exploitation. The answer likely depends on implementation details—interest rates, fee structures, transparency, recourse mechanisms, and genuine commitment to serving rather than extracting from marginalized communities. Technology alone is neutral; the ethical valence depends entirely on how it is deployed and governed.

4.4 Toward Humane AI-Driven Financial Inclusion

Achieving genuinely humane and inclusive AI-driven financial services requires multifaceted interventions across technical, regulatory, and institutional domains:

Technical interventions can address fairness, transparency, and privacy challenges through several approaches that make AI systems more accountable.

Making algorithms transparent rather than black-box requires explainability techniques. Often, when AI denies credit or increases insurance premiums, users receive little explanation. Counterfactual explanations flip this paradigm by showing exactly what would need to change: "If your mobile payment consistency increased from 60% to 80%, you would qualify for this loan." This specificity transforms opaque rejections into actionable guidance. Similarly, quantifying each factor's contribution to decisions—"location contributed 10%, transaction history 45%, digital footprint 35%, with 10% from other factors"—enables users to understand why decisions were made and effectively contest errors.

Preserving privacy while enabling learning presents a fundamental tension in AI-driven financial inclusion: alternative data improves credit scoring but raises surveillance concerns. Privacy-preserving techniques like federated learning resolve this tension by keeping sensitive data on users' phones or at local institutions—the AI model travels to where data resides, learns patterns locally, and only shares aggregated statistical updates rather than raw information. Multiple microfinance institutions can collaboratively improve credit models without ever sharing customer databases. Differential privacy adds carefully calibrated statistical noise ensuring no individual's information can be extracted from trained models while preserving overall accuracy.

Proactively identifying when models fail requires adversarial testing—systematically challenging AI systems with edge cases before deployment. Developers feed models applications from unusual occupations, transaction patterns from remote rural areas, and data combinations rarely seen during training. This stress-testing reveals failure modes where models make egregiously wrong decisions for specific subpopulations, allowing fixes before vulnerable users are harmed. For instance, adversarial testing might reveal that a credit model systematically underscores seasonal agricultural workers, prompting recalibration.

Regulatory frameworks must evolve beyond traditional financial regulation to address algorithmic accountability. Requirements might include mandatory algorithmic impact assessments before deployment, regular fairness audits testing for disparate impacts across demographic groups, explainability standards requiring comprehensible decision explanations, data governance frameworks limiting alternative data collection and use, and accessible appeals processes for algorithmic decisions. The challenge lies in crafting regulation that protects vulnerable populations without stifling beneficial innovation—adaptive governance rather than rigid control.

Institutional commitments ultimately determine whether AI serves inclusion or exploitation. Financial service providers must adopt fiduciary mindsets, designing products that genuinely serve client welfare rather than merely extracting fees. This requires performance metrics that measure actual improvements in client financial wellbeing—emergency savings accumulated, debts repaid, insurance claims paid, wealth built—not just transaction volumes or revenue generated. Participatory design processes should involve intended beneficiaries in product development rather than imposing technocratic solutions. Human oversight must complement automation, preserving agency and providing recourse when algorithms fail.

Cross-sector collaboration among technology providers, financial institutions, regulators, civil society organizations, and affected communities is essential to navigate the complex tradeoffs inherent in AI-driven financial inclusion. No single actor possesses sufficient expertise or perspective to ensure humane deployment. Multi-stakeholder governance mechanisms can facilitate ongoing dialogue, surfacing concerns and adapting practices as understanding evolves.

The evidence from Global South implementations—from M-Pesa's transformation of financial access in Kenya to challenges with digital savings platforms in Africa, from alternative credit scoring's expansion of lending across multiple markets to concerns about predatory practices—demonstrates both AI's transformative potential and its capacity to create new forms of exclusion and exploitation when deployed without adequate safeguards. Success requires unwavering commitment to centering human dignity, social justice, and genuine inclusion rather than merely technological sophistication or financial efficiency.

5. Conclusion

Artificial intelligence emerges as a transformative enabler of financial inclusion, operating powerfully across payments, savings, lending, insurance, and investments through innovative applications of supervised learning, unsupervised learning, reinforcement learning, natural language processing, computer vision, and generative AI. Real-world implementations across the Global South—from M-Pesa's AI-enhanced identity verification in Kenya to Nubank's automated investment services in Brazil, from Tala's alternative credit scoring across multiple African markets to BIMA's micro-insurance serving tens of millions, from India's UPI ecosystem's multilingual chatbots to Nigeria's AI-powered agricultural insurance—demonstrate AI's potential to democratize financial services at unprecedented scale. By integrating evidence from contemporary global research and practical deployments, this analysis illustrates how AI fundamentally transforms access, equity, and financial stability for populations previously marginalized from formal financial systems.

Credit invisibility represents 20% of the adult population in developed economies and even higher proportions in developing markets (Consumer Financial Protection Bureau, 2015). The technology's capacity to process alternative data, automate complex decisions, personalize services at scale, and operate efficiently in resource-constrained environments promises unprecedented opportunities for economic empowerment across diverse global populations. Research demonstrates that alternative data models achieve comparable or superior predictive power to traditional credit scores while dramatically expanding access—mobile payment data correlates with creditworthiness at 0.65-0.72, matching FICO score performance (Björkegren & Grissen, 2019). AI-driven microfinance reduces operational costs from 6-12% to under 2%, enabling profitable lending at scales previously impossible (Milana & Ashta, 2021). Alternative data scoring overwhelmingly relies on gradient boosted tree methods like XGBoost and LightGBM, which power 70-80% of production systems because they automatically detect non-linear relationships and this can reduce costs by 6% to 25% of total losses (Khandani et al., 2010).

However, realizing this transformative potential requires unwavering commitment to responsible deployment, robust ethical oversight, and sustained efforts to mitigate critical risks including algorithmic bias that perpetuates discrimination, privacy violations through invasive data collection, digital exclusion of populations lacking connectivity or devices, opaque decision-making that prevents meaningful contestation of automated decisions, and predatory practices disguised as financial inclusion (Solon Barocas & Selbst, 2016; Kalluri, 2020). Alternative data can encode existing societal inequalities when machine learning trains on data reflecting historical discrimination, creating proxy discrimination even without explicitly using protected characteristics (Solon Barocas & Selbst, 2016). Mission drift represents a profound risk in microfinance, where algorithmic optimization of repayment prediction can gradually shift portfolios away from the poorest populations toward easier-to-serve segments, abandoning original social missions (A. Ashta & Hudon, 2012; Cull, Demirgüç-Kunt, & Morduch, 2011). The inclusion versus exploitation dilemma emerges in the interest rate paradox: AI enables lending to previously excluded populations, but often at 30-40% APR or higher: rates that enable access while potentially trapping borrowers in debt spirals (Rona-Tas & Guseva, 2018).

Success demands that financial inclusion efforts prioritize human dignity over efficiency metrics, ensure transparent governance with meaningful accountability mechanisms, proactively address digital divides through investment in infrastructure and digital literacy, maintain accessible human oversight and appeals processes, and rigorously evaluate whether AI deployments genuinely improve financial wellbeing or merely extract value from vulnerable populations (Kalluri, 2020; Kroll et al., 2017). Cultural context must inform AI systems—recognizing that irregular income flows may reflect seasonal agriculture rather than financial instability, and that family financial support represents cultural values rather than poor financial discipline (Anderson et al., 2009; Clark, 1997). Financial behavior doesn't translate uniformly across cultures, and AI models trained on Western credit bureau data may see culturally appropriate patterns as high risk. Fairness constraints must balance predictive accuracy with equitable outcomes across demographic groups, accepting minor performance reductions to achieve demographic parity and prevent discriminatory impacts (S. Barocas et al., 2019). Regulatory frameworks increasingly recognize that algorithmic neutrality doesn't guarantee fair outcomes, requiring explainability, disparate impact testing, and fairness-aware machine learning approaches (Bracke et al., 2019; V. Bumacov et al., 2014).

Only through such conscientious approaches—informed by voices from affected communities, guided by ethical frameworks that center human rights and social justice, regulated by governance structures that hold technology providers accountable, and continuously evaluated against outcomes for the most marginalized—can AI genuinely advance financial inclusion while honoring principles of justice, respecting human dignity, and fostering sustainable, equitable development that genuinely serves historically underserved communities rather than creating new forms of technological exploitation or algorithmic exclusion (Bélanger et al., 2025; Kalluri, 2020). The future of AI-driven financial inclusion depends not merely on technological sophistication, but on our collective commitment to deploying these powerful tools responsibly, ethically, and in genuine service to those who need them most. Ethical AI-driven financial inclusion requires preserving human agency, aligning objectives with borrower welfare, involving borrowers in design, auditing relentlessly for bias, respecting data dignity, and maintaining epistemic humility—recognizing that algorithms trained on thousands of loans know patterns while experienced loan officers know context, and both are valuable (Kalluri, 2020).

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Brief bio

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